

The administration of aromatics at Haghia Triada: two classes of oil, a block of ingredients, and the format of the spice lists

Alberto Acedo Biome Makers Inc. Draft v0.2, 9 September 2026. The identification of *303 is measured and declared as disputed; the Assur comparison is made from Escobar's edition. Not for circulation.

Abstract

The oil logograms of Linear A fall into two classes by the quantities that follow them and by the company they keep: bulk oils (OLE+U, plain OLE, OLE+KI, OLE+MI, OLE+DI, OLE+TU) are recorded in whole units and share their documents with grain, olives and figs; qualified oils (OLE+TA, OLE+RI, OLE+NE) are recorded in fractions and share their documents with the logogram *303, conventionally read as cyperus, and with a closed block of four further signs (*308, MI+JA+RU and the ligatures QA₂+[]+PU and QA₂+[]+RE), all above chance under hypergeometric tests on 1,719 documents. Those documents share a format otherwise absent from the Linear A tablets: single signs used as commodity abbreviations followed by small fractions, which is the format of the Mycenae Ge spice records. Read by quantity, the dossier has the structure of a Bronze Age unguent formula: within each tablet the secondary ingredients share a single fraction ($p < 0.00005$ against the corpus distribution of fractions, across two sites and two hands), and the dominant ingredient carries between three and five times each of them. That structure is the one attested in the Middle Assyrian perfume recipes from Assur, in the kyphi recipe of Dioscorides and in the Pylos Un deliveries to the perfumer. Read by function, cyperus is not the fragrance but the astringent of the *stypsis*, the treatment that makes oil able to hold a scent and which Theophrastus says is particularly necessary with olive oil; the astringent-to-oil ratio in HT 23a (0.67) falls within the range Dioscorides prescribes (0.55-1.03). This explains an anomaly of the archive: cyperus occurs 84 times in 61 Linear A documents and is the third commodity of the record, against three occurrences in the far larger Mycenaean corpus and two in Middle Assyrian. Crete was not importing an exotic spice: it was growing the reagent that its own olive oil required.

Keywords: Linear A, aromatics, perfumed oil, cyperus, stypsis, Haghia Triada, Mycenae Ge series, administrative format.

1. Introduction

The Mycenaean perfume industry is well documented: the Pylos Fr tablets record oil qualified by its aromatic (ku-pa-ro-we, of cyperus; wo-do-we, of rose; pa-ko-we, of sage), the Un tablets assign the ingredients to the perfumer (coriander, cyperus, root, wine, honey, wool), and the Mycenae Ge tablets list condiments per person with the smallest measures of the system (Sarpaki 2001; Shelmerdine 1985). On the Minoan side, the literature has described the oil logograms and the aromatic logograms separately (Palmer 1995; Schoep 2002; Del Frio, Nosch and Rougemont 2010) without asking whether they form a dossier. This paper asks that question with the

methods described in Acedo (2026c): association between signs by document tested against the hypergeometric distribution, document format, and control by scribal hand.

1 bis. The identification of *303, which is disputed, and is measured here

The reading of *303 as cyperus, which this paper follows from the edition consulted, is not unanimous. Goold and Pope (1955), followed by Uchitel (2003), identify it as barley; Palmer (1995) proposed emmer wheat. All three identifications rest on the form of the sign, on its relative frequency, or on the comparison of resources across archives, and none on its quantitative distribution within the corpus. That distribution is worth measuring, since the premise conditions everything that follows.

sign	n	median	maximum	records with a fraction
GRA (grain, identified by inheritance from Linear B)	90	15.62	976	13%
olives	21	10.00	93.5	29%
wine	34	3.62	104	29%
oil	100	3.62	100	26%
figs	50	2.00	75	34%
*303	**64**	**0.62**	**23**	**61%**

Most records of *303 are for less than one unit and six in ten carry a fraction, against the grain of the same archive, whose median is twenty-five times greater and which records fractions in one case in eight. The company differs in the same way: *303 does not co-occur with olives in any of its 53 documents, while grain does so in twelve of seventy-two; and only four documents carry both signs, which rules out their being varieties of one product.

The control lies in Linear B, where the identification is not in doubt: there grain reaches 10,300 units and cyperus reaches ten. The profile of *303 (median 0.62, maximum 23, majority in fractions) is the second and not the first.

This does not demonstrate that *303 is cyperus. It demonstrates that **it is not counted as a cereal**, which rules out the most cited alternative reading and leaves the field to substances dosed in fractions: an aromatic plant, a spice or a resin. It also answers the frequency argument, the principal one in favour of a cereal: *303 can be the most recorded commodity at Khania without being a grain, if Khania was a centre of aromatics production, which is what the chemistry of the region's workshops indicates. And the possibility of ritual barley does not account for sixteenth fractions, which are dosage rather than offering, nor for mixture in fixed proportion with oil.

One datum tells against this reading and should be recorded. In the four tablets combining a personnel section with a commodity section (HT 27a, HT 89, HT 94a, HT 100), the commodity section always follows the same order, *303 followed by figs and wine, with additions at the end; this is the sequence Uchitel (2003) calls Standard II. That *303 systematically heads that sequence places it in the position of the principal product, which is what a cereal would occupy in a ration list. Position alone does not decide, but it is an argument for the alternative reading and not for the quantitative profile set out above.

Everything that follows holds under any identification compatible with that profile, and the measurements in this paper (the shared fractions, the proportions, the associations) are independent of it.

The block measured with two nulls

The hypergeometric test that isolates this block assumes documents are exchangeable, and in this corpus they are not: measured on material with no effect, it reports a false positive in 8.0% of trials at the 5% threshold, and 11.9% when both signs come from a single site. The cause is the site structure. The measurement is therefore repeated with a null that preserves it, a bipartite configuration model that swaps elements only between documents of the same site and whose false-positive rate, measured the same way, is 5.5%.

member	documents with *303	hypergeometric p	p with site fixed
figs	17	< 0.00001	< 0.002
*308	4	0.00006	< 0.002
MI+JA+RU	4	0.00006	0.0033
OLE+NE	2	0.00566	0.0017
wine	11	< 0.00001	0.0117
CYP+D	4	0.00016	0.0167
CYP+E	4	0.00016	**0.0533**
OLE+RI	2	0.00566	**0.0517**

Six of the eight members keep the association when site is held fixed. Two, CYP+E and OLE+RI, fall above the threshold and are declared as such: their association with *303 cannot be distinguished from what sharing a site would produce. The reason shows in their distribution, since CYP+E occurs in eight documents at Khania and two at Haghia Triada, and the seed itself is mostly from Khania, whereas *308 and MI+JA+RU divide between the two sites in proportions unlike its own. Both values are in any case very close to the threshold, and what should be claimed is not that the association is false but that the corpus does not allow it to be told apart from the effect of site. Two claims are worth keeping apart, since they are often confused. That two signs occur in the same documents is co-occurrence, measured as above; that

one immediately follows the other is adjacency, a different and stronger claim. Measured against a null that shuffles the elements within each document, and therefore preserves which signs share a document while destroying only their order, *303 is followed by figs fifteen times against 5.6 expected, and figs by wine sixteen against 4.1, while grain followed by figs does not exceed the null. The sequence the literature describes as recurrent is therefore real adjacency and not an effect of co-occurrence. The instrument, calibrated on those pairs of the corpus that share enough documents and with their order destroyed, reports a false positive in 2.5% of trials at the 5% threshold and 0.25% at the 0.1% threshold.

2. Oil falls into two classes

The oil logograms of Linear A fall into two groups by their quantities and by their company. OLE+U, plain OLE, OLE+KI, OLE+MI, OLE+DI and OLE+TU are recorded in whole units (medians between 3 and 7, up to 54) and share documents with grain, olives, figs and *304, above chance in every case (hypergeometric tests on 1,719 documents): they are the oils of farm production. OLE+TA, OLE+RI and OLE+NE are recorded in fractions and share documents with *303 (OLE+TA in 4 of 7 documents, $p < 0.001$; OLE+RI and OLE+NE in 2 of 4 each, $p = 0.006$), with *308, MI+JA+RU and the ligatures QA₂+[]+PU and QA₂+[]+RE, and almost never with grain or olives: HT 23a, HT 30, HT 32, HT 35, HT 60, KH 61 and KH 85 belong to this group, and on HT 30 and HT 100 the allocation heading SA-RA₂ is followed by *303 with OLE+TA. The second group corresponds in function to the qualified oils of the Pylos Fr series (ku-pa-ro-we, wo-do-we, pa-ko-we) and to the condiment, perfume and dye plants that Sarpaki (2001) gathers from the Mycenaean texts and the archaeobotanical record, with the qualifier written as a ligatured sign instead of as a word — a device Linear B also uses in CYP+KU and PYC+KU, and which the editors of the new ivory inscriptions from the Knossos cult centre describe in the same terms, syllabograms ligatured to vessel logograms as acrophonic qualifiers of type or content (Kanta, Nakassis, Palaima and Perna 2024). The values of TA, RI and NE are unknown, and OLE+TA, the one most often recorded with *303, is the natural candidate for cyperus-scented oil.

3. The block of ingredients

The four signs that accompany the qualified oils form a closed block. Over the 1,719 documents of the corpus, *308 (8 documents), MI+JA+RU (7), QA₂+[]+PU (5) and QA₂+[]+RE (5) co-occur with the logogram *303 and with OLE+TA, OLE+RI and OLE+NE far above expectation (hypergeometric, $p < 0.001$ in every case), are always recorded in fractions, and *308 and MI+JA+RU never appear with grain or figs. The two QA₂ ligatures also occur under SA-RA₂ allocation headings. A fifth sign that a first pass had included, *188, does not belong to the block: it is recorded in whole units, heads documents and appears with KI-RO deficits, so it is another commodity and is removed from the dossier. The documents of the block are HT 23a, HT 30, HT 32, HT 35, HT 60, HT 100, KH 61 and KH 85, to which HT 16 and HT 20 are added by format (section 4). One limitation of this test must be declared. The documents containing the block are richer in logograms than the corpus average (4.0 against 1.4), and that richness inflates any co-occurrence: the hypergeometric test controls how many documents each sign occurs in,

but not how many logograms each document carries (capture-artefact control, $r = 0.415$). What sustains the result is therefore not raw co-occurrence but two things that document richness does not explain: that the fixed-margin null, which does preserve per-document sums, produces the same grouping; and that the contrast between what the block accompanies (oil and olives, 7 documents against 1.9 expected, $p = 0.002$) and what it never accompanies (textiles, zero of fourteen) is asymmetric, whereas richness would affect both equally. The grouping of these documents is likewise not new. Uchitel (2003) classifies the Haghia Triada tablets into standard sequences of commodities, and his Standard Sequence III gathers HT 23a, HT 32, HT 33 and HT 35, that is, four of the five central tablets of this dossier, on the different criterion of which ideogram combinations are present. In the same work he identifies SA-SA-ME as the Minoan equivalent of Mycenaean sa-sa-ma, “sesame”, with the reference to MY Ge 602; that identification is taken here from him and not offered as our own. What this paper adds to the grouping is its measurement: the hypergeometric association of each sign with the block, the equality of fraction among secondary entries within each tablet, the exclusion of a textile reading, and the ratio between the astringent and the oils.

4. The format: Minoan spice lists

HT 16 deserves separate mention. Its entries are single signs used as commodity abbreviations followed by small fractions (WA $\frac{1}{3}$, *188+KU $\frac{3}{8}$, SA-PO ZE $\frac{1}{16}$), a format found among the Linear A tablets only in HT 20 as well (WA $\frac{1}{4}$, *188+KU $\frac{1}{2}$), written by the same hand as HT 16 according to Godart’s attribution (GORILA V, scribe HT 10), so that the format is in part a scribal habit (on the scope of palaeography as a criterion, Salgarella and Judson 2024); HT 30 and HT 32 are by scribe HT 1, and HT 23a, HT 35 and HT 60 have no attribution. It is the format of the Mycenae Ge series, in which condiments are recorded by acrophonic abbreviation with the smallest measures of the system.

5. Sesame in this dossier

SA-SA-ME (AB 31-31-13) is attested once, on HT 23a, with the fraction $\frac{1}{16}$, in the same type of document that supports the other cases: a record of aromatics with the logogram *303, three qualified oils (OLE+NE, OLE+TU, OLE+RI), wine and entries in minimal fractions. In the Mycenae Ge series sesame, sa-sa-ma, occupies that same position, that of the smallest measures of the system, beside fennel, safflower and mint. Greek σήσαμον is a recognised Semitic loan (Akkadian šamaššammū, Ugaritic ššmn; Masson 1967; Rosół 2013), and the Minoan form preserves the reduplication of the source with a final vowel of its own, which places it as a link of the Minoan channel that Melena and Piquero (2015) posit for commodity terms.

The case belongs to another class than the names formed on plant words, and this is stated: it is not a name formed on a plant’s name but the commodity word itself, transmitted from the East to Greek through Minoan. Its force is smaller: a single attestation, without a family, without affixes and with no possible co-occurrence null; and the ambiguity between commodity and party, which the seals decided for KU-PA, cannot be decided here (a sign-group followed by a fraction is the shape of an entry and also of an allocation to someone, as WA-DU-NI-MI 3 $\frac{1}{6}$ in the fruit

list of HT 6b). What tips the balance towards the commodity is the format of the document, the absence of derivatives and of any occurrence in a heading, roundel or personnel list, and the exact parallel of the Ge series.

6. The external frame: the documented cyperus recipe

Two things established by historical botany and by Mycenology place this dossier and had not been brought into relation with it.

The first is the plant. Negbi (1992) separates the two sedges of the ancient Mediterranean: *Cyperus esculentus*, the tiger nut, a sweetmeat plant whose tubers were eaten in Egypt boiled in beer or ground with honey and are found in tombs from the fourth millennium; and *Cyperus rotundus*, the perfume plant, whose tubers were used in perfumery and medicine by Egyptians, Mycenaeans and Greeks, and which Theophrastus, Pliny and Dioscorides record. If *303 is cyperus, as the conventional reading holds, it is the second.

The second is the formula. Dioscorides describes an Egyptian preparation of *Cyperus* tubers with other plant ingredients, honey and wine, and the same preparation appears in the Ebers papyrus, so that the recipe is documented over sixteen hundred years (Negbi 1992). Compared with HT 23a the structural coincidence is notable: cyperus first and with the largest quantity among the fractions, several minor plant ingredients in equal parts, and wine with one of the large quantities (VIN 10). The missing element is honey, and that absence is a prediction rather than a problem: if the Minoan formula belongs to the same tradition, one of the HT 23a signs with a large quantity should be it. The hypothesis identifies none of them and is left formulated as such.

On the Mycenaean side, the reference study of cyperus in the archives is Melena (1974) on ku-pa-ro in the Knossos tablets, and Foster (1977) describes at Knossos an administrative department devoted to perfumery and offerings, which is exactly the kind of unit this dossier appears to document two centuries earlier at Haghia Triada.

7. The structure of a recipe, measured and compared

The dossier does not merely group the same signs: it groups them in the same numerical shape, and that shape is the shape of a Bronze Age perfume recipe.

The secondary ingredients share a fraction within each tablet. With the fraction letters of GORILA and the values of the optimal system of Corazza, Ferrara, Montecchi, Tamburini and Valério (2021), HT 23a has cyperus at B (1/5) and five consecutive entries at H (1/16): *308, three qualified oils and MI+JA+RU. HT 35 has cyperus at 1 and three entries at B (1/5). HT 32 has three entries at unity. KH 12 has cyperus at 2 and two entries at J (1/2). The probability that all four tablets display that homogeneity when fractions are drawn from the corpus distribution is below 0.00005 (in twenty thousand simulations it never occurs; the mean is 0.41 homogeneous tablets). Two sites and two hands are involved, so it is not a scribal habit.

And the principal aromatic carries between three and five times each secondary: 3.2 to 1 on HT 23a, 5 to 1 on HT 35, 4 to 1 on KH 12. The values of the fraction letters are in dispute, and the conflict bears on this calculation, so it should be declared. Montecchi (2009) shows that the

only tablet permitting a certain conclusion with no prior corrections or emendations is HT 104, and derives $J = \frac{1}{2}$ and $E = \frac{1}{4}$ from it, with $B = \frac{1}{5}$ from the proportion $10:4 = J:B$; for K she hesitates between $\frac{1}{10}$, on formal resemblance to the Mycenaean submultiples, and $\frac{1}{16}$, on the geometric progression of HT Zd 156, and reports that the first option conflicts with that result. The optimal system of Corazza, Ferrara, Montecchi, Tamburini and Valério (2021) gives $K = \frac{1}{10}$ and $D = \frac{1}{6}$. With $K = \frac{1}{10}$ the two HT 23a entries carrying that letter (SA-SA-ME and QI-RI-TU-QA) stand at 1.6 parts, at an intermediate level between cyperus and the base; with $K = \frac{1}{16}$ they stand at 1 part, that is, at the level of the base, and the structure of the tablet goes from three levels to two. The figures that follow use the 2021 system; with the other, the astringent-to-oil ratio does not change, since neither quantity involved carries K.

That structure is the one of the best documented contemporary perfume recipes, and the comparison can be made with figures. Of the seven perfumery fragments from the Middle Assyrian archive at Assur published by Ebeling (1950), only two are recipes proper, PKT 6 (VAT 9659) and PKT 7 (VAT 9493), identified by the conditional heading *šumma i ša [X] turaqqa*, “if you process oil of X”; of the remaining five, only KAR 220 carries an equivalent heading, referring to cane oil (Song 2025). KAR 220 is moreover the recipe signed by the perfumer Tappūti-Bēlat-ēkalle, and its modern edition, used here, is Escobar’s (ORACC Glass project; Escobar 2023).

The recipes have six elements: a conditional heading, a vehicle with its quantity, a principal ingredient in large quantity, secondaries in quantities equal to one another, water measured separately, and the procedure with numbered pourings.

The feature that matters here is the fourth, and in KAR 220 it can be counted. Reduced to litres on Escobar’s equivalences ($1 \text{ qû} \approx 1 \text{ litre}$, $1 \text{ sūtu} \approx 10 \text{ litres}$), the recipe distributes fifteen entries over seven distinct values: 40 litres of steeped liquid mixture and 20 of cane with its roots as the dominant quantities; **six distinct ingredients at exactly 3 litres** (crushed nut-sedge, myrtle, crushed cane, cane, myrrh and sweet myrrh); a single intermediate of $1\frac{1}{2}$ litres (almond); **three ingredients at exactly 1 litre** (himimu, iaruttu and myrtle); and four traces at $\frac{1}{2}$ and $\frac{2}{5}$.

Set beside it, HT 23a distributes fourteen entries over six values. On the fraction values of the system the consulted edition uses ($D = \frac{1}{3}$, $K = \frac{1}{16}$): one entry at 17 (E), four at 10 (wine, QA_2+RE , $QA_2+RE+PU$ and $SI+ME$), one at 1 (KO-RU), a single one at $\frac{1}{3}$ (KA-NA with cyperus), **five sharing $\approx \frac{1}{6}$** (*308, OLE+NE, OLE+TU, OLE+RI and MI+JA+RU) and **two sharing $\frac{1}{16}$** (QI-RI-TU-QA and SA-SA-ME). On the values of Corazza et al. ($D = \frac{1}{6}$, $K = \frac{1}{10}$), the same signs give $\frac{1}{6}$, $\approx \frac{1}{12}$ and $\frac{1}{10}$ respectively, so that **the distribution into tiers is identical and only the magnitudes change**: one entry alone at the intermediate level, five sharing the next and two sharing the lowest. What the argument requires is that coincidence between distinct entries, and it does not depend on the system. Two qualifications about these figures, which reading Corazza et al. (2021) makes necessary. The first is that the edition consulted **resolves sign combinations to their value**: where the tablet carries two D signs, the transcription gives a single value, so a transcribed quantity may correspond to more than one

sign. Across the corpus there are 285 isolated fractions and only three consecutive pairs, which confirms the practice. The second is that the five entries sharing a fraction **do not carry an exact value but one the edition marks as approximate**, a mark that appears 28 times in the corpus and never absent. What the argument requires, however, is not the value: it is that the five entries carry **the same sign** and therefore the same quantity, whatever it is. That its value is disputed between $1/5$ and $1/6$, and that the edition marks it as approximate, does not affect the coincidence; if anything it strengthens it, since it shows the coincidence does not rest on a contested assignment. A third control must be applied to this reading, and Younger himself supplies it. His continuity principle holds that in many tablets the last name or commodity cited governs until another situation is stated, so that an entry without an element of its own falls under the preceding one. Measured over the corpus, that principle governs 47% of the runs (156 of 331 series of logograms under a single word have two or more members). If HT 23a applied it, its five entries sharing a fraction would not be five ingredients but five quantities of one. **It does not apply it: each of its fourteen entries carries its own element and its own quantity**, with not a single governed run. In a corpus where nearly half the series economise on naming the product, this tablet names it fourteen times, which is proper to a formula and not to a list of allocations. The shape is the same: a dominant that fixes the volume of the operation, an intermediate occupied by a single element, and two levels shared by groups of secondaries. **That several distinct ingredients should carry exactly the same quantity is what distinguishes a formula from an inventory**, since in a record of stock there is no reason for such coincidence, whereas in a recipe the proportion is precisely what is being noted.

It is worth adding that nut-sedge figures among the six 3-litre ingredients of KAR 220, alongside myrtle and cane. And that the scale of these operations was modest: Tappūti's recipe produces twenty litres, and a head perfumer might supervise the processing of up to a hundred (Escobar 2023), so that the order of magnitude of a Bronze Age perfumery operation is tens of litres. Five of those six elements are present in HT 23a. The one missing is the procedure, and its absence is systematic in Linear A, as are the absence of a verb of transfer and of a date (Acedo 2026g).

The dossier is not a dyeing dossier. Over the 1,719 documents of the corpus, 67 carry something from the block and 14 carry textile logograms: the intersection is zero, against 0.5 expected by chance; and the block coincides with oil and olives in 7 documents against 1.9 ($p = 0.002$). The Middle Assyrian archive, which has both industries, keeps its dyeing vocabulary (argamannu, takiltu, hûratu, gabû) entirely separate from its perfumery vocabulary. That exclusion, which up to this point rests only on the absence of textile logograms, has a second and independent support in residue analysis. Gas chromatography of the Minoan dye workshop at Alatsomouri-Pefka, of Middle Minoan IIB date, identifies murex purple, yellow from weld, red from madder and **lanolin**, the grease of sheep's wool (Koh et al. 2016). The profile of a Minoan perfume workshop is disjoint from it: olive oil as base, storax resin, camphor, linalool from coriander and manoyl oxide from the Cretan rock rose (Koh and Birney 2017). The two lists share not one substance, and the wool marker that defines the dye workshop is precisely the one

absent both from the analysed perfumed product and from this dossier, which co-occurs with textiles in none of its documents.

The behaviour of the signs allows a class to be assigned, not a name. *308 is the only member of the block that occurs outside it, accompanying olives on HT 123+124a with values from 26% to 28% in four sections: it is a product of the olive at a fixed yield, not an aromatic. MI+JA+RU is the only one that does NOT associate with foodstuffs above chance (2 of 8 documents, $p = 0.12$) and the one that most often appears only with aromatics (6 of 8): it has the profile of a perfumery aromatic, which in Assur are the ones measured in shekels. QA₂+PU is the opposite: six of its eight documents carry foodstuffs (expected 0.6; $p < 0.0001$) and three carry wine ($p = 0.0009$), which is the position of an ingredient at once alimentary and perfumatory. QA₂+RE does not separate from QA₂+PU and shares its base sign, so the two are plausibly varieties of the same thing. And three further signs associate with the dossier without having been identified before: *306 (four of its six documents, $p < 0.0001$, exclusive to Khania), *305 ($p = 0.0065$, exclusive to Haghia Triada) and *307 ($p = 0.0009$).

A point of usage about those three signs is called for, and the palaeographic study of Ferrara, Montecchi and Valério (2022) supplies it: *306 and *310 are among the signs used in syllabic sequences and without a hypothetical phonetic transcription, whereas *307 is among those that behave like commodity logograms or accompany them. Checked in the corpus, *306 occurs sixteen times inside sequences and seven on its own, so it is above all a sign that forms words; its association with this dossier may be due to those words occurring in these documents rather than to its designating a commodity. It is therefore recorded here as an associated sign and not as a candidate ingredient. To this may be added that Uchitel has proposed, in a paper devoted to the question, that *306 is the Linear A sign for “woman”, on the basis of its occurrence with numbered recipients against agricultural commodities and of the pairing of roundels and tablets in which it occupies the position of the sign DI. If that reading is correct, *306 falls outside this dossier by a second route, independent of the first. And *305, which also associates, in fact belongs to a neighbouring dossier described below.

The same study also answers a question this work left open. Of the twenty-two Linear A signs attested ten or more times that have NO counterpart in Cretan Hieroglyphic, five are the central ones of this dossier: *302 (oil), *303 (cyperus), *306, *307 and *308. The authors propose that these signs were introduced, or became frequent, in Linear A in response to changing economic or administrative practices, which would account for the absence of equivalents in the hieroglyphic material. **The industry of aromatic preparations is therefore organised with the script that records it and not before it**, and that palaeographic result coincides with two independent arguments: that the system of commodity ideograms with quantities is an innovation of Linear A, and that the form of the Minoan receipt is native and not adopted (Acedo 2026g).

The words of the dossier. The two main recipes are headed by sign-groups belonging to the family studied in the companion paper: KA-NA heads HT 23a, and KA-KU-PA heads the spice list HT 16. On HT 20, a list with the format of the Mycenae Ge series, KU-MA-JU and SA-RE-

JU appear with small fractions. And TI-TI-KU, which heads the recipe HT 35, reappears closing the votive inscription ZA Zb 3 from Zakros: it is one of the very few words in the corpus that cross the boundary between accounting and cult, and the only one of them that survives a detailed examination.

What is not reached. The name of any of these substances. The acrophonic route, which works in Linear B and is demonstrated precisely for cyperus (CYP+KU and PYC+KU = ku-pa-ro, as OLE+PA = pa-ko-we and GRA+PE = pe-ma), yields no candidate here: of the twenty perfume substances documented in the Bronze Age, none begins with pu- in Greek or in Akkadian, and the Assyrian balsam is per'aduhhu and not re-. Moreover, and this is a fact about the system, the ligatures of the Minoan dossier abbreviate words that the corpus never writes out in full, unlike Linear B, where ku-pa-ro is written as well as ligatured.

8. The formula read by function: cyperus as astringent

The classical sources describe manufacture in TWO STAGES (Shelmerdine 1985, ch. 2, with Dioscorides I.42-63 and Theophrastus, *De odoribus* 14-18). The first, the *stypsis*, does not scent: it prepares the oil by adding weakly scented astringents, and the three the sources name are aspalathus, **cyperus** and ginger-grass. That treatment makes the oil receptive to the fragrances that follow and incidentally thickens it; the astringents are kneaded with wine or water into a paste and then heated in the oil. Theophrastus adds the point that matters here: the treatment **is particularly necessary with olive oil, which does not retain odours well**. In the second stage the treated oil is given its final fragrance by cold maceration.

With that frame, the roles in HT 23a order themselves by their quantities: cyperus in the largest fraction is the astringent; the three qualified oils and *308, in the smallest fraction, the vehicle being prepared; wine and the two QA₂ ligatures, in whole units, the kneading liquids, which the sources enumerate as wine, mead and their mixture (Theophrastus 47); and sesame, QI-RI-TU-QA and MI+JA+RU the aromatics of the following stage.

And that frame makes possible a numerical check that would not otherwise exist. The ratio of astringent to oil is the quantity that defines the *stypsis*. In Dioscorides (I.56) it runs from 0.55 to 1.03 according to the recipe. In HT 23a, cyperus ($1/5 = 0.200$) over the three qualified oils ($3 \times 1/16 = 0.1875$) gives 1.07, and over those three plus *308 (total $1/4$) gives 0.67. **Both fall within the range.** The comparison is legitimate, unlike those that cross corpora with different units, because both quantities are on the same fraction scale on the same tablet. And it reinforces by a second independent route that *308 is a fatty product: the ratio fits better when it is counted as oil.

This also explains an economic fact that the archive documents and that has hitherto had no cause. Cyperus occurs 84 times in 61 Linear A documents and is the third most recorded commodity of Crete, ahead of grain and of olives; in the Mycenaean archives it occurs three times and in Middle Assyrian twice, despite those being larger corpora. **Crete was not consuming cyperus: it was producing it at scale.** And the reason is in Theophrastus: Crete is

an olive-oil country, and olive oil is precisely the oil that most needs an astringent. An unguent industry based on olive oil consumes cyperus industrially; one based on the Egyptian *balanos* oil, “the least viscous”, scarcely needs it. Cyperus was not the fragrance of Crete: it was the reagent that made it possible to scent its oil.

On the side of agrarian economy, cyperus co-occurs with figs (17 documents, expected 1.6; $p < 0.0001$) and with wine (11, expected 1.1; $p < 0.0001$) and **never with olives or with livestock**: it belongs to the irrigated accounting and not to the dry-farming one, which suits a plant of damp soils (Dioscorides: “it grows in clayey or marshy places”).

9. The missing slot and the candidates that are ruled out

Comparing six formulae by functional slot (fatty vehicle, astringent, kneading liquid, intermediate level, final aromatic, resin or fixative), HT 23a has five filled and one empty: **resin or balsam appears in all the others and is not identified here**. In the kyphi it is 2.5 kg of resin; in Dioscorides’ iris perfume, 1.7 kg of balsam-wood; in KAR 140+ from Assur, 12 qû of balsam and later 3; and in the Chamalevri workshop the residue analysis identifies resin and beeswax. The candidates by position are the two hapax of the tablet, QI-RI-TU-QA (1/10) and SI+ME (10 units), and with a single attestation each no decision is possible. The Cretan resin par excellence does have a name in the corpus: KI-TA appears on HT 29, KI-TA-I on HT 123+124a beside *308 and the olives, and KI-TA-NI-TE on a vessel from Palaikastro. One archaeobotanical datum favours that candidacy: in the Artisans’ Quarter at Mochlos, where a Late Minoan IB aromatics workshop has been identified, the study of the plant remains records the collection of *Pistacia*, probably for its resin, at a site where the olive is moreover the most abundant plant of the assemblage (Sarpaki, in Soles and Davaras 2004).

One further slot, which Theophrastus documents and which is worth opening: the **colourant**. “Some perfumes are given a colour [...] the dye used for colouring red perfumes is alkanet; sweet marjoram-perfume is dyed with the substance called *khroma*, a root imported from Syria” (*De odoribus* 31). By its nature it goes in minimal quantity, and the only entry on HT 23a with quantity 1, in final position, is KO-RU.

Several candidates are ruled out with evidence. **Water** cannot be present: there is no water logogram in Linear A or in Linear B, because what has no value does not enter an account; that the Assyrian recipes measure it does not contradict this, since they are workshop instructions and not inventories. **Coriander** does not fit the intermediate level either: on PY Un 267 it goes in a quantity equal to cyperus (six and six), so on HT 23a it should stand at the astringent’s level and not at half of it. **Rose** has no formal candidate in the corpus, and Theophrastus warns that flower perfumes last two months and spoil within the year: what does not keep is not stored and not inventoried, which is the same reason the archive is full of roots, resins and oils. **Iris**, identified chemically at Chamalevri, is the best functional candidate for the final aromatic, but has no Mycenaean word or logogram to compare with; and it is worth noting that *wi-ri-za*, which accompanies cyperus on PY Un 249, is ρίζα “root” and not iris. And **saffron**, the emblematic plant of Minoan iconography, has a logogram in Linear B (CROC, 43 attestations) and none in

Linear A; the quantities of the three unidentified signs of the dossier (*305 reaches 90, *307 reaches 156, *306 ranges from 1 to 8) fit none of them with a spice that costs a hundred and fifty thousand flowers per kilogram.

10. The direct Mycenaean parallel

The Pylos Un tablets record the delivery of ingredients to the perfumer, which is the same genre of document. PY Un 267 reads: “Aksotas delivered to Thuestas the perfumer, aromatics for boiling the unguent: coriander 6, cyperus 6, *157 16, fruit 2 T 5, wine 20, honey 2, wool 2.” Five of its seven elements coincide with HT 23a, and the two that do not are the verb of delivery, the office of the recipient and the purpose, none of which Crete ever writes. The tablet confirms in a readable language two things inferred here: that the aromatics are delivered in quantities equal to one another, and that the liquid goes in a much larger quantity than the rest. And it offers a lesson of scale: even in a language that is read, with the office and the purpose written out, *157 remains unidentified. That three of the fourteen elements of HT 23a remain unidentified is not a limitation of the procedure but the normal condition of this genre of document.

11. Discussion

What is described is an administration of aromatics and scented oils: a class of oil measured in fractions and allocated together with cyperus and a fixed set of ingredients, in documents with a format of their own, in which the allocation heading SA-RA₂ appears twice followed by cyperus and OLE+TA. The Mycenaean, two centuries later, write the same thing with words: the oil is called “of cyperus”, “of rose”, “of sage”. The Minoans wrote it with ligatures, and that is why the dossier had not been seen: it is not in the vocabulary but in the signs attached to the logograms. The editors of the Knossos ivory inscriptions describe the ligatures of their vessels in the same terms, acrophonic syllabograms specifying type or content (Kanta, Nakassis, Palaima and Perna 2024), which supports the reading.

A necessary correction of category. In the ancient tradition, perfume, medicine and offering are not separate things: Dioscorides describes cyperus as diuretic, good for the womb, for the chest and for ulcerated mouths, and “effective for thickening unguents”, and he says of the kyphi that it is a temple perfume, an ingredient of antidotes and a draught for asthmatics. His work is entitled *De materia medica* and its first book is entitled “Aromatics”. What this dossier discriminates is dyeing; it does not discriminate between perfumery, pharmacy and cult, because in the Bronze Age those were made with the same ingredients, in the same proportions and probably in the same workshop. “Aromatic preparations” is therefore the correct label, and the distinction between cosmetic, medicinal and ritual is ours and not the sources’.

What cannot be decided. First, the values of TA, RI and NE: they are acrophonic qualifiers of something, and the corpus does not say of what; OLE+TA, which occurs with *303 in four of its seven documents, is the candidate for “cyperus oil”, and that is a hypothesis, not a reading. Second, the identity of *308, MI+JA+RU and the two QA₂ ligatures: their distribution places them in the dossier and no more. Third, whether the proportions constitute a canonical recipe: the

ratios of principal to secondary run from 3.2 to 5, a narrow range but not a single figure, so the formula is one of composition and not of fixed dosage.

The Minoan channel, measured over a whole semantic field. The hypothesis that certain commodity terms reached Greek from the East through Minoan (Melena 1974; Melena and Piquero 2015) has been argued word by word. With the list of twenty Bronze Age perfume substances, closed by the Assur recipes and the Mycenaean series, it can be measured in the aggregate: 90% of their Greek names are non-Indo-European (18 of 20), against 35% of a control set of twenty concrete nouns from other domains (7 of 20; Fisher's exact test, $p = 0.0008$). And within the non-Indo-European group, the Pre-Greek ones outnumber the Semitic two to one: κύπειρος, μύρτος, κορίαννον, σφάκος, τέρμινθος, μίνθη, μάραθον, σέλινον, κνῆκος, ρόδον and κηρός against μύρρα, βάλσαμον, σήσαμον, κύμινον and ἄμωμον. When the Greeks inherited the perfume industry they inherited its lexicon too, and that lexicon was not theirs and did not for the most part come directly from the East. The qualification is necessary: "Pre-Greek" is not equivalent to "Minoan", since it designates the Aegean substrate in general; what points to Minoan specifically is that six of those substances have counterparts attested in Linear A itself (the KU-PA family, mi-ta, ka-na, sa-sa-me, ku-mi-na-qe and ki-ta).

This dossier is not the accounting of Haghia Triada but one of its accountings, and it is worth situating. Montecchi's revised classification (2019) creates a class Ω gathering seven tablets around the sign *305, one of those associated here with the block without having been identified before; seeding the same association analysis with it yields a second block of six members (*305, *318, ZU, TU, QA₂+RE+PU and OLE+NE) which shares only six of its twenty-four documents with the sixty-eight of the first. The two are distinguished above all by how they count: the aromatics dossier records 54% of its quantities in fractions and the *305 dossier only 5%. And by their company: the first associates with foodstuffs and with oil ($p < 0.001$ for both) and the second with people ($p = 0.011$), besides carrying the double mina, the only identifiable weight unit of the corpus, which does not appear here. The contact between them lies in wine and in the QA₂ family, which distributes three ligatures of the same base sign between the two sectors, with the compound form QA₂+RE+PU in the second. That comparison reinforces the description above, since it shows that the properties of the aromatics dossier (fractions, oil, foodstuffs) are not those of the archive in general but its own.

What this dossier contributes to other discussions. It gives context to the family of sign-groups that accompanies the logogram *303 and is analysed elsewhere (Acedo 2026a); it reinforces the identification of *303 as cyperus, which is conventional (Palmer 1995; Schoep 2002), by showing that the sign lives in an aromatics dossier and not in a cereal one, and by showing that the quantity it carries is the quantity an astringent carries; and it places the Pylos perfume industry as a later, mainland development that should not be projected onto Minoan usage.

What would refute this work

This work must state, under the third requirement of the protocol its author proposes, what observation would refute it. Three would. First, that a residue analysis of process vessels from a

Minoan aromatics workshop should find no *Cyperus* markers (cyperene, α -cyperone, cyperotundone, rotundene) while finding those of the other proposed ingredients: reading cyperus as an astringent predicts that they be there and not in vessels of finished product, and their absence from both leaves the reading unsupported. Second, that a new tablet should record *303 in large whole quantities alongside olives or livestock, which would place it in the profile of a cereal rather than of a dosed substance. And third, that the equality of fraction among secondary entries should turn out to be an artefact of how fractions are transcribed rather than a property of the documents, which a systematic treatment of the fraction readings in the primary edition could demonstrate.

Data

Transliterations: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer (Hogan 2022), with the hand attributions of GORILA V; sign verification in SigLA (Salgarella and Castellan 2020). Assyrian recipes: TCMA (Ebeling 1950). Tests: hypergeometric over documents and permutation nulls following Acedo (2026c); scripts in the kuro package (Acedo 2026d).

Declaration of assistance

Corpus processing, the statistical tests and the first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

References

- Acedo, A. 2026a. Minoan names formed on plant words. Draft.
- Acedo, A. 2026c. What can be claimed about a corpus of seven thousand signs. Draft.
- Acedo, A. 2026g. The Minoan receipt is native: the form of bookkeeping in six Bronze Age administrations. Draft.
- Corazza, M., S. Ferrara, B. Montecchi, F. Tamburini and M. Valério 2021. The mathematical values of fraction signs in the Linear A script. *Journal of Archaeological Science* 125, 105214.
- Del Frio, M., M.-L. Nosch and F. Rougemont 2010. The terminology of textiles in the Linear B tablets. In C. Michel and M.-L. Nosch (eds), *Textile Terminologies in the Ancient Near East and Mediterranean*, Oxford, 338-373.
- Dioscorides. *De materia medica* I.42-63, ed. R. T. Gunther, 1934.
- Escobar, E. A. 2023. Tappūtī-bēlat-ekalle (fl. 1200 BCE): a cuneiform tablet on Middle Assyrian perfumery. In *Women in the History of Science*, UCL Press, 15-21.
- Song, Y. 2025. Perfume-making and craft-knowledge transmission in the Middle Assyrian period. *'Atiqot* 118, 99-112.
- Ebeling, E. 1950. *Parfümrezepte und kultische Texte aus Assur*. Rome: Pontificium Institutum Biblicum.

- Foster, E. D. H. 1977. An administrative department at Knossos concerned with perfumery and offerings. *Minos* 16, 19-51.
- Godart, L. and J.-P. Olivier 1976-1985. *Recueil des inscriptions en linéaire A, I-V*. Paris.
- Hogan, R. 2022. *Linear A Explorer*. <https://lineara.xyz>
- Kanta, A., D. Nakassis, T. Palaima and M. Perna 2024. *Kadmos* 63.
- Koh, A. J. and K. J. Birney 2017. Organic compounds and cultural continuity: the Penn Museum Late Minoan IIIC stirrup jar from Tournioti. *Mediterranean Archaeology and Archaeometry* 17(2), 19-33.
- Masson, E. 1967. *Recherches sur les plus anciens emprunts sémitiques en grec*. Paris.
- Koh, A. J., P. P. Betancourt, M. N. Pareja, T. M. Brogan y V. Apostolakou 2016. Organic residue analysis of pottery from the dye workshop at Alatsomouri-Pefka, Crete. *Journal of Archaeological Science: Reports* 7, 536-538.
- Koh, A. J. y K. J. Birney 2017. Organic compounds and cultural continuity: the Penn Museum Late Minoan IIIC stirrup jar from Tournioti. *Mediterranean Archaeology and Archaeometry* 17(2), 19-33.
- Melena, J. L. 1974. Ku-pa-ro en las tablillas de Cnoso. *Emerita* 42, 307-336.
- Melena, J. L. and J. P. Piquero 2015. Minoan channel words in Mycenaean Greek.
- Montecchi, B. 2009. Le frazioni, gli errori di calcolo e le unità di misura nella documentazione in lineare A. *Annali dell'Istituto Italiano di Numismatica* 55, 29-52.
- Goold, G. P. y M. Pope 1955. *The Cretan Linear A script*.
- Uchitel, A. 2003. Local differences in arrangements of ration lists on Minoan Crete. En M. Brosius (ed.), *Ancient Archives and Archival Traditions*, Oxford, 143-150.
- Negbi, M. 1992. A sweetmeat plant, a perfume plant and their weedy relatives: a chapter in the history of *Cyperus esculentus* L. and *C. rotundus* L. *Economic Botany* 46(1), 64-71.
- Palmer, R. 1995. Linear A commodities: a comparison of resources. In R. Laffineur and W.-D. Niemeier (eds), *Politeia*, Liège, 133-155.
- Rosół, R. 2013. *Frühe semitische Lehnwörter im Griechischen*. Frankfurt.
- Salgarella, E. and S. Castellan 2020. *SigLA*. <https://sigla.phis.me>
- Salgarella, E. and A. Judson 2024. *Palaeography as a chronological criterion*.
- Sarpaki, A. 2001. Condiments, perfume and dye plants in Linear B. In A. Michailidou (ed.), *Manufacture and Measurement*, Athens, 195-265.
- Schoep, I. 2002. *The Administration of Neopalatial Crete*. Salamanca.

Shelmerdine, C. W. 1985. The Perfume Industry of Mycenaean Pylos. SIMA Pocket-book 34, Göteborg.

Steele, P. M. and T. Meißner 2017. From Linear B to Linear A: the problem of the backward projection of sound values. In P. M. Steele (ed.), *Understanding Relations Between Scripts*, Oxford, 93-110.

Salgarella, E. 2020. Aegean Linear Script(s). *Rethinking the Relationship between Linear A and Linear B*. Cambridge.

Soles, J. S. and C. Davaras (eds.) 2004. Mochlos IC. Period III. Neopalatial Settlement on the Coast: The Artisans' Quarter and the Farmhouse at Chalinomouri. *The Small Finds. Prehistory Monographs 9*. Philadelphia: INSTAP Academic Press.

Duhoux, Y. 1978. Une analyse linguistique du linéaire A. In Y. Duhoux (ed.), *Études minoennes I*, Leuven, 65-129.

Theophrastus. Concerning Odours (*De odoribus*), in *Enquiry into Plants*, ed. A. Hort, Loeb Classical Library, 1916, vol. II.

Tzedakis, Y. and H. Martlew (eds.) 1999. *Minoans and Mycenaeans: Flavours of their Time*. Athens.

Weilhartner, J. 2014. The influence of Aegean iconography on the design of the Linear B logograms for animals, plants and agricultural products. In G. Touchais et al. (eds), *Physis, Aegaeum 37*, Leuven and Liège, 297-304.

Weingarten, J. 1987. Seal-use at LM IB Ayia Triada: a Minoan elite in action. I. Administrative considerations. *Kadmos 26*, 1-43.

Wiener, M. H. 2015. The Mycenaean conquest of Minoan Crete. In *The Great Islands: Studies of Crete and Cyprus presented to Gerald Cadogan*, Athens, 131-142.

Jiménez Delgado, J. M. 2026. Personal communication, 8 September 2026.

Sommerschild, T., Y. Assael, J. Pavlopoulos, V. Stefanak, A. Senior, C. Dyer, J. Bodel, J. Prag, I. Androutsopoulos and N. de Freitas 2023. Machine Learning for Ancient Languages: A Survey. *Computational Linguistics 49*, 1-44.

Younger, J. G. 2010. The Myrtos-Pyrgos and Gournia roundels inscribed in Linear A: suffixes, prefixes, and a journey to Syme. In *Cretan Offerings. Studies in Honour of Peter Warren*, London, 383-388.

Younger, J. G. 2024a. Linear A Lexicon (updated 26 May 2024). <https://www.academia.edu/120182223>

Younger, J. G. 2024b. Linear A Texts in Phonetic Transcription: Haghia Triada; Other Texts. Formerly at people.ku.edu/~jyounger/LinearA/; republished on Academia.edu, 2024.